CP–6284

Sub. Code
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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

ENGLISH AND COMMUNICATION SKILLS

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Read the passage carefully and answer the questions below : (8)

One winter day in 1972, a Danish tanker truck loaded with phenol, a corrosive acidic compound, spurn out of control and turned over. Its poisonous Cargo spread into a stream, killing fish and jeopardized the water supply of a nearby village. After clearning up the spill, govt officials hammered out new safety measures for the transport of dangerous substances.

(i) When did the accident happen?

(ii) What is phenol?

(iii) What was the nature of accident?

(iv) What was the impact of the accident?

(b) Write an essay on an Indian Marriage. (7)

Or

(c) Write a precis of the following passage : (8)

Another way of looking at the situation is to do a SWOT analysis. SWOT stands for strengths, weaknesses, opportunities, threats. Make a heading for each word and start identifying the appropriate items in each category. Ideally, you should do this with a group of people, using a large flipchart of wallboard. Let the items come up in random, allocating them where they belong. Lets look at our Granby example :

Strengths :

(i) High-quality, good looking aeroplane

(ii) Strongly built-excellent safety record

(iii) Very efficient (18 mpg at 180 mph)

(iv) Fart, good, long-range performance

(v) Porsche usage, great heritage

(vi) Fun to fly

(vii) Available now, no warting list

(viii) Granby owners love the curcraft

(ix) Strong, well-run owner's club.

(d) What are the modalities to be followed in precis writing? (7)

2. (a) Write an essay on the objectives of written communication. (8)

(b) Write a letter to your trend about your visit to a historical place. (7)

Or

(c) What are the importance of commercial letters and explain its kinds. (8)

(d) Write a formal invitation to your friends and relatives inviting them to your son's first birthday. (7)

3. (a) You are the special correspondent of 'The Hindu' send a news report on our prime ministers recent visit to Australia. (8)

(b) Write an application for the post of a clerk in the PWD in Tamilnadu. (7)

Or

(c) Write a letter to the editor of a reputed newspaper expressing your views on 'compulsory voting' to influence the public. (8)

(d) A major train accident occurred between Chennai and Hydrabad. Collect information from eye-witnesses and others and send a detailed report to the local newspaper. (7)

4. (a) What are the good communication skills? (8)

(b) What are the importance of communication skills.(7)

Or

(c) What are the five key elements of active listening? (8)

(d) What are the barriers to effective human communication? (7)

5. (a) What are the types of interviews? (8)

(b) Write a vote of thanks to a chief guest of your association function. (7)

Or

(c) Define five tips to deliver the best extempore speech. (8)

(d) Write an essay on the communication skills in the work place. (7)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

NAUTICAL MATHEMATICS – I

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions.

$(5 \times 20 = 100)$

1. (a) (i) Show that the points $(1,3,1)$, $(1,1,-1)$ $(-1,1,1)$ and $(2,2,-1)$ are lying on the same plane. (6)

(ii) For any vector $\overline{a}$, prove that $i \times (\overline{a} \times i) + \overline{j} \times (\overline{a} \times \overline{j}) + \overline{k} x (\overline{a} \times E) = 2\overline{a}$. (6)

(iii) A company has two plants to manufacture scooters. Plat II manufacturing 80 percent of the scooters and plant II manufacture 20 percent. At plant I 85 out of 100 scooters are rated standard quality or better. At plant II, only 65 out of 100 scooters rated standard quality or better. What is the probability that the scooter selected at random came from plant I if it is known that scooter is of standard quality. (8)

Or

(b) (i) Find the coefficient of correlation between industrial production and export using the following data :

Production (x) :	54	55	56	59	60	60	62
Export (y) :	35	38	37	39	44	43	44

(ii) From the following data relate to the marks of 10 students in the internal test and the university examinations.

Internal marks :	25	28	30	32	35	36	38	39	42	45
University marks :	20	26	29	30	25	18	26	36	35	46

Obtain the two regression equations and determine the most likely internal marks for the university mark of 25.

2. (a) (i) Find the vertex, focus, diretrix, equation of the latus rectum, length of latus rectum for the parabola $y^2 - 8x + 6y + 9 = 0$. (10)

(ii) The orbit of the earth around the sun is elliptical in shape with sun at a focus. The semi-major axis is of length 92.9 million miles and centricity is 0.017. Find how close the earth gets to sun and the greatest possible distance between the earth and the sun. (10)

Or

(b) (i) An iron cone of radius 9cm and height 28cm is melted and spherical iron balls of radius 3 cm is made. Find the number of iron balls made. (7)

(ii) A river is 40 m width. The depth d in meter at a distance x meters from one bank is given by the following data :

x :	0	5	10	15	20	25	30	35	40
y :	0	7	6	8	7	6	4	3	0

Use Simpson's rule to estimate the area of cross-section of the river. (13)

3. (a) (i) Explain Napier's rule for right-angled spherical triangles and quadrant spherical triangles. (7)

(ii) Solve the spherical triangle, given $A = 129^\circ 5' 28''$, $B = 142^\circ 12' 42''$, $C = 105^\circ 8' 10''$ $5' 28''$. (13)

Or

(b) (i) Explain Haversine formula and its advantages over sine and cosine formula . (8)

(ii) Solve the spherical triangle, given $a = 37^\circ 48' 12''$, $b = 59^\circ 44' 16''$, $C = 90^\circ$. (12)

4. (a) (i) If $y = x^{i^{x^{x^{\cdot^{\cdot^{\infty}}}}}}$, prove that $\frac{dy}{dx} = \frac{y^2}{x[1 - y \log x]}$. (6)

(ii) Differentiate $Sin^{-1}(3x - 4x^3)$ with respect to x. (8)

(iii) If $y = ae^{2x} + be^{-x}$, show that $y_2 - y_1 - 2y = 0$. (6)

Or

(b) (i) Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$. (6)

(ii) P.T $\int_0^{\pi} \theta \sin^3 \theta \, d\theta = \frac{2\pi}{3}$. (6)

(iii) Find the area bounded by one arch of the curve $y = \sin ax$ and the x-axis. (8)

5. (a) (i) Reduce the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 5 \\ 3 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ to canonical form. (10)

(ii) Solve the following system of equations $x + y + z = 2$, $2x - y + z = 5$, $2x - z = 0$. (10)

Or

(b) (i) Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 3 \\ 1 & 1 & 2 \end{pmatrix}$. (8)

(ii) Find the eigen values and eigen vector of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$. (12)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

NAUTICAL PHYSICS AND ELECTRONICS — I

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Explain center of gravity with some examples. (3)

(b) State and explain Newton's experimental law of collision. (4)

(c) Describe an experiment of determine the surface tension of a liquid by capillary tube method. (8)

Or

(d) State and explain Kepler's law. (4)

(e) Obtain an expression for loss of kinetic due to oblique impact of two smooth spheres. (8)

(f) Explain the following: (3)

(i) Time of flight

(ii) Range.

2. (a) Define battery? Write its characteristics. (3)

(b) What is R.M.S and average value? Derive expression for each interms of maximum value. (7)

(c) Briefly explain the construction and works of dip circle. (5)

Or

(d) Explain about electric lighting circuits. (4)

(e) Give the theory of transformer. (6)

(f) Write a short note on magnetic elements of the earth. (5)

3. (a) Discuss briefly the main applications of Bernoulli's principles. (6)

(b) Obtain dimensional Stoke's law. (3)

(c) Derive the construction and working of Bourdon pressure geuge.

Or

(d) Distinguish between streamlined and turbulent flow. (4)

(e) Describe the Poiseuille's method for the determination of coefficient of viscosity of a liquid. (6)

(f) Explain construction and working of marine hydrometer. (5)

4. (a) Explain photometer. (4)

(b) Discuss the diffraction due to single slit. (5)

(c) Discuss the construction and uses of optical pyrometer. (6)

Or

(d) Obtain Einstein's photoelectric equation. (4)

(e) Discuss the working of photo diode with neat diagram. (5)

(f) Describe the construction and working of an astronomical telescope. (6)

5. (a) Explain the fabrication of N-type semi conductor. (3)

(b) Describe the construction, working of a bridge rectifies with neat diagram. (7)

(c) What is thermister? Explain its characteristics. (5)

Or

(d) Explain the action of PN junction diode. (5)

(e) Explain how zener diode acts as a voltage regulator and percentage of regulation. (5)

(f) Describe the construction and characteristic of an PNP transistor. (5)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

NAVIGATION — I

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. Define the following nautical terms :

   (a) (i) Pole
   (ii) Axis
   (iii) Meridians
   (iv) Latitude
   (v) Longitude.

   Or

   (b) (i) Mean latitude
   (ii) Geographical mile
   (iii) Nautical mile
   (iv) Equator
   (v) D'Long.

2. (a) Find the course and distance using parallel bailing, for the given data :

From	Lat 40°24' N	Long 086°38' E
To	Lat 40°24' N	Long 093°41' E

Or

(b) Find the RHUMB Line course and distance from starting position (P) 10°19' N 140°40' W to arrival position (Q) 02°12° S 160°18' W.

3. (a) Ship 'A', in lat 30° S steers due west at 15 kts. Ship 'B' in lat 22° S also steers due west. They commenced from the same longitude. If after 20 hrs. they remained due North, and south of each other. Calculate speed of 'B'.

Or

(b) A vessel in position 45°15' N/29°22' E steers 090(T) at 15 kts. After 8 hours, her position was found to be 45°15' N/27° E. Find the set and drift.

4. Find the initial course, final course and GC distance in the following cases :

(a) From 49°50' N/005°15' W

To 32°29' N/064°00'.0 W

Or

(b) From 19°00.0 N/067°00.0 W

To 00.0 N/006°00.0 W

5. (a) Describe the SOLAR system.

Or

(b) State the laws of "Planetary motion" by Kepler.

(c) What are inferior and superior planets.

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CP–6288

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15

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

SHIP OPERATION TECHNOLOGY – I

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) (i) Explain the hazards associated in Tankers. (5)

(ii) How the temperature for the cargo maintained in Reefer type of vessels? (5)

(iii) What do you mean by Main Engine Control room and the Navigation Bridge and where do you find them? (5)

Or

(b) Explain with suitable sketches the following : (Each 3 Marks)

(i) DB Tanks

(ii) Monkey Island

(iii) Fo'c'sle Deck

(iv) Cargo Holds

(v) Navigation Lights.

2. (a) What do you understand by :

(i) Cock a Bill

(ii) Anchor Aweigh

(iii) All made fast Fore and Aft

(iv) Pilot Away

(v) Courtesy Flag.

Or

(b) What basic knowledge is required to carryout the duties as a "look out"?

(c) Define risk assessment and safe working practice.

3. (a) Explain the purpose of a look-out and the COLREG Rule associated with it. (15)

Or

(b) (i) State how many points in a Mariner's Compass. (5)

(ii) List out the safety wears of sailor's on board.(5)

(iii) State how many types of paint defects. (5)

4. (a) What is corrosion and state the measures taken to prevent it on board? (15)

Or

(b) State the types of ropes and wires used and where they are used. (15)

5. (a) Define :

(i) Mechanical Advantage (8)

(ii) Efficiency Rate. (7)

Or

(b) Explain how will you secure the hatches with respect to an expected heavy weather. (15)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

VOYAGE PLANNING, COLLISION PREVENTION AND MARINE COMMUNICATION – I

Time : 2 Hours Maximum : 25 Marks

Answer **all** questions.

($5 \times 5 = 25$)

1. (a) (i) List out any eight informations available on a navigational chart. (2)

(ii) Compare two different types of chart projections. (3)

Or

(b) Write short notes on : (1 + 1 + 1 + 2)

(i) Chart Datum

(ii) Deviation

(iii) Variation

(iv) Natural scale.

2. (a) (i) What are different categories of charts. (2)

(ii) Why the nearest Latitude scale is used for measuring distance? (2)

(iii) Name the authorities publishing charts in INDIA and U.K. (1)

Or

(b) (i) What is meant by "Fixing the Ship"? Mention any six ways of fixing the vessel. (3)

(ii) Explain the following : (2)

(1) Leading lights.

(2) Sailing around the arc.

3. (a) Given the following, calculate True and Compass Bearing : (5)

(M) Bearing of object : 280°

Variation : 3° (E)

Deviation : 2°(W)

Or

(b) (i) Draw a representation of ships deviation card. (3)

(ii) State three methods of check Gyro Error. (2)

4. (a) Describe the following. Draw suitable diagrams :

(i) DR position

(ii) Observed position

(iii) Leeway. (2 + 2 + 1)

Or

(b) (i) Define SET and DRIFT. How are they handled in navigation? (3)

(ii) What is E.P? (2)

5. (a) Define the following as per COLREGS :

(i) The word-vessel

(ii) Sailing vessel

(iii) Underway

(iv) NUC

(v) CBD.

Or

(b) Define :

(i) Stand on vessel

(ii) Give way vessel

(iii) Overtaking vessel

(iv) Fishing vessel. (2)

(c) State the rule regarding "Safe Speed". (2)

(d) How will you join or leave a T.S.S? (1)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Nautical Science

NAVIGATION III

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) (i) Explain great circle sailing.

(ii) Find great circle distance, initial course and final course from

(1) 06° 00 N 79° 00 W

To

(2) 38° 00 S 179° 00 E. (15)

Or

(b) Find the initial course, final course and the distance along the composite track from

(i) 51° 20' N 010° 00 E

To

(ii) 52° 00 W 0.55° 00 E

having a ceiling latitude of 53°N. (15)

2. (a) Explain the following :

(i) Sidereal period of the moon.

(ii) Moon's nodes.

(iii) Synodic period of the moon. (15)

Or

(b) Explain Lunar eclipse and the types of Lunar eclipse with the aid of a diagram. (15)

3. (a) (i) Define circumpolar bodies. State the conditions necessary for a body to be circumpolar.

(ii) Why do we calculate the Azimuth of a body? (15)

Or

(b) Explain the following nautical terms :

(i) LHA

(ii) SHA

(iii) Rational horizon

(iv) Azimuth

(v) Amplitude. (15)

4. (a) On 30th April 2008, in DR position 00° 20'N 60° 12' W, the sextant altitude of the sun's u.c. east of the meridian was 44° 13'.4 when chron (error 03^m 09^s slow). Showed 00^h 57^m 43^s. If I.E. was 3'.1 off the ABC and H.E. was 20 mtr., find the direction of the P.L. and the longitude where it crosses the D.R. Lat. (15)

Or

(b) On 6th March 2008, in EP 52° 12'N, 170° 40'E, the sextant altitude of the sun's U.L. near the meridian was 31° 59'.8 when chron (error 01^m 50^s fast) showed 01^h 29^m 20^s. If I.E. was 2.3 on the ARC and H.E. was 40 mtr, find the direction of the P.L. and the position through which to draw it. (15)

5. (a) Explain the principle of an ECHO sounder. Draw a block diagram and explain the controls. (15)

Or

(b) (i) Explain the properties of free gyroscope.

(ii) Explain gyroscopic TILT and DRIFT. (15)

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32

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

SHIP OPERATION TECHNOLOGY – III

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Mention any five physical characteristics to be considered while manouvering A ship. (5)

(b) What preparations are made while making port? (Cover w.r.t. papers, cargo works, safety)? (10)

Or

(c) What do you mean by manouering? (2)

(d) Using a sketch, explain the emergency manouere to recover a man over board. (8)

(e) Explain turning circle and circle. (5)

2. (a) Mention any five types of anchors. And discuss abut the widely used one. (8)

(b) Why should you report the state of cable while dropping and weighing anchor? (7)

Or

(c) Define the following : (15)

(i) Walk back the anchor

(ii) Anchor a cock-bill

(iii) Foul anchor

(iv) Dredging of anchor

(v) Bitter and why is it so called.

3. (a) Differentiate between current and tide. (5)

(b) How do the above factors affect ship handling? (5)

(c) Berthing/unberthing is a team work. Discuss. (5)

Or

(d) Enumerate the environmental factors influencing ship handling. (5)

(e) Ship handling is both art and science? Discuss. (10)

4. (a) What measures may be adapted to prevent damage to ships in heavy weather? (8)

(b) What dangerous conditions may be encountered in bad weather? (7)

Or

(c) What special precautions should be observed for launching of life boat in rough weather? (10)

(d) Briefly discuss the influence of wave-length on different sizes of ships. (5)

5. (a) Inspection and maintenance are inclusive to each other. Discuss. (10)

(b) How does the preparation's differ while docking in graving dock vis A vis A floating dock? (5)

Or

(c) Why is it necessary for a vessel to stay in class? (5)

(d) Define survey. (5)

(e) What is docking plan? (5)

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33

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

VOYAGE PLANNING, COLLISION PREVENTION AND MARINE COMMUNICATION – III

Time : 2 Hours Maximum : 25 Marks

Answer **all** questions.

(5 × 5 = 25)

1. (a) (i) Write about the introduction of GMDSS, replacing old system.

(ii) What are GMDSS areas?

Or

(b) (i) What different identification numbers are assigned to a vessel?

(ii) Write short notes on Radio spectrum.

2. (a) (i) What equipments are fitted onboard which are related to weather?

(ii) What is the responsibility of a ship regarding cyclone warning?

Or

(b) (i) What is the special communication arrangements during SAR operation?

(ii) How many flags are maintained onboard? Draw and write the meaning of frequently used flags?

3. (a) (i) Describe the importance of keeping VHF16 watch onboard.

(ii) Write a "securitae" message.

Or

(b) (i) What different log books are maintained onboard and how are they important?

(ii) How with you treat a distress alert message?

4. (a) (i) State Rule 5, lock out as per COL REG.

(ii) How important is a look out during sailing?

Or

(b) (i) How would you like the use the RADAR during different phases of your voyage?

(ii) What actions are recommended to avoid collision?

5. (a) Mention different types of Buoys. White short notes on list of lights.

Or

(b) (i) Write short notes on "Guide to port entry".

(ii) What is your contingency plan to determine the position of your u/c when your GPS fails?

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

NAVAL ARCHITECTURE – I

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions. (5 × 20 = 100)

1. (a) Draw and explain the following parts of a ship.

   (i) Steering gear compartment

   (ii) Hatch covers of a cargo hold.

Or

   (b) Define the following :

   (i) Prismatic coefficient

   (ii) Block coefficient

   (iii) TPC

   (iv) DWA

   (v) Reserve Buoyancy.

2. (a) A Box shaped vessel 18 × 5 × 12 m^3 floats in sw at a draft of 6.1 m. Calculate the RB%.

Or

   (b) A box shaped vessel 120 m long and 15 m wide has life draft of 4 m and load draft of 9.8 m is SW. Find light displacement, load displacement and dead weight.

3. (a) Sketch and explain the three states of Equilibrium.

Or

(b) The transverse cross sectional areas of a ship's under water portion 90 m long are 0.5, 22.9, 49, 73.5, 88.5, 8.3, 58.6, 31.8, 14.2, 8.1, 4.5 m^2.

The last given area is at the after perpendicular of the ship. The spacing between last three sections is half the common interval between the rest. Find the AB and displacement in SW.

4. (a) Define the following with sketches :

(i) LBP

(ii) Tumble Home

(iii) Rise of Floor

(iv) Dead weight

(v) Light weight.

Or

(b) Write short notes on the following :

(i) Precautions while welding

(ii) Edge preparation

(iii) Various test on welded joints.

5. (a) Explain in detail about the NDT methods carried out to know the quality of welding.

Or

(b) A vessel has two deep tanks, P and S each 12 m long, 5 m wide and 8 m deep. The port side is full of SW while the starboard side is empty $w = 9840\,t$. kM = 8.5 m, kG = 8 m. Find the GM fluid if SW is transferred from P to S until each tank has equal quantity of ballast.

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35

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

MARINE ENGINEERING AND CONTROL SYSTEMS – I

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

1. (a) Explain the following properties of material :

   (i) Ductility

   (ii) Malleability

   (iii) Hardness. (5)

   (b) A simple supported beam of 10 meter long, supported at extreme ends carries concentrated load of 20 kN, 40 kN, and 50 kN at a distance of 3 meter, 6 meter and 8 meter respectively from the left end. Draw the 'Shear force' and 'Bending Moment' diagram. (10)

Or

   (c) Explain : (5)

   (i) 'Scalar' and 'Vector' quantity.

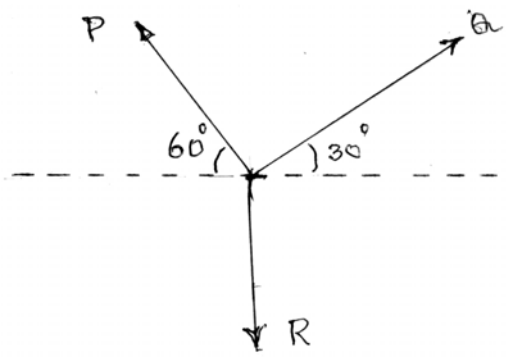
   (ii) Mass and weight.

   (d) Define : (5)

   (i) LAMI's Theorem.

   (ii) Newton's laws of motion.

(e) Three forces P, Q and R are acting on a particle as shown in the figure and is keeping the particle in equilibrium. If the force R is 500 kN, find the other two forces P and Q. (5)



2. (a) Explain the following : (4)

(i) Laminar Flow

(ii) Turbulent Flow.

(b) Sketch and describe 'Hydraulic press'. (4)

(c) A fresh water pipe line tapers uniformly from 120 mm diameter at the bottom to 60 mm diameter at the top, the difference in height being 6 meter. When the discharge quantity is 0.0424 m^3/ sec and the pressure at the bottom is 160 kN/m^2. Find the pressure at top. (7)

Or

(d) Define :

(i) Specific volume

(ii) Specific heat. (4)

(e) Explain the law of gravitation. (4)

(f) 0.5 m^3 of perfect gas at a pressure of 85 kN/m^2 and temperature at 16°C are compressed to a volume of 0.13 m^3 and the final pressure is 560 kN/m^2. Calculate the final temperature. (7)

3. (a) Define: (6)

(i) First Law of Thermodynamics.

(ii) Second Law of Thermodynamics

(iii) Zeroth Law.

(b) Define the terms : (4)

(i) Density

(ii) Relative density.

(c) In an isothermal compression process, air at an initial pressure of 200 kN/m^2 and volume 3 m^3 respectively compressed to 1200 kN/m^2. Find the final Volume. (5)

Or

(d) Define the terms : (4)

(i) Latent heat

(ii) Sensible heat.

(e) Explain the following : (6)

(i) Saturated steam

(ii) Dry steam

(iii) Superheated steam.

(f) What are the Boiler mountings? (5)

4. (a) Explain : (4)

(i) Ohm's Law

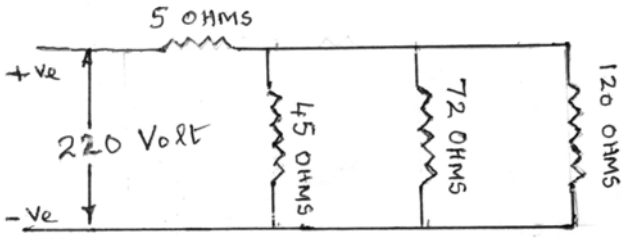
(ii) RMS value.

(b) What are the conditions to be fulfilled for taking on load in parallel with

(i) An incoming Alternator (AC) with a Running alternator on load.

(ii) An Incoming Generator (DC) with a Running Generator (DC) on load. (5)

(c) Find the current in each Resistor. (6)



Or

(d) Define : (4)

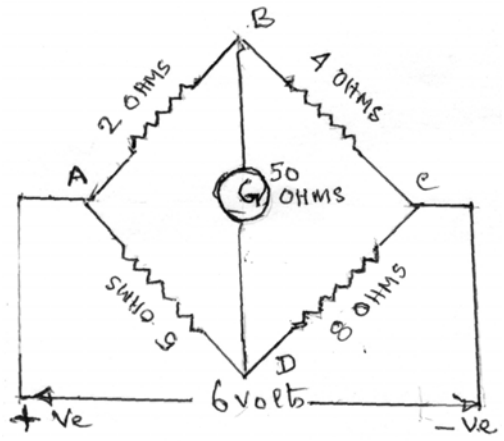
(i) Joules Law

(ii) Kirchoff's Current Law.

(e) Explain magnetic effect of AC and DC electric current. (4)

(f) A Wheatstone bridge network is arranged as shown below. Find the current and its direction in the galvanometer circuit.

Note: Ignore the internal resistance of the 6 volt battery. (7)



5. (a) Explain various drive for Main propulsion system. (3)

(b) Give details of 'Auxiliary Machinery' on board ship. (4)

(c) What is the use of Sewage treatment plant and when it is used?

Explain the operation of a Sewage treatment system you have studied. (8)

Or

(d) Explain hydraulic steering gear with hunting valve function. (8)

(e) What are the various Fyel oil and lube oil tanks in the machinery space. (4)

(f) Explain the use of Purifiers on board. (3)

CP–6295

Sub. Code
36

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

ENVIRONMENTAL STUDIES

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions.

(5 × 20 = 100)

1. (a) Environmental Science is multi-disciplinary in Nature — Discuss. (10)

   (b) Explain the structure of atmosphere. (10)

   Or

   (c) State in detail the scope and importance of environmental science. (10)

   (d) Suggest your measures to maintain the Public awareness about the environmental Protection. (10)

2. (a) Mention any six natural resources explain the impact of depletion of any two of them. (10)

   (b) What are the effects of in discriminate Mining? (10)

   Or

   (c) What is the best solution to meet the energy requirement of the world? (10)

   (d) Science and technology are gradually Overcoming the lack of Natural resources – Discuss. (10)

3. (a) How does the population growth affect the eco system? (10)

(b) How does the environmental science contribute towards the well being and survival of humanity? (10)

Or

(c) Mention any five environmental problems and what is the solution. (10)

(d) What is greenhouse effect? State the greenhouse gases and indicate their sources. (10)

4. (a) What are the major reasons for urbanization? What measures can be taken to discourage the same? (10)

(b) What is the role of IT in creating environments consciousness? (10)

Or

(c) What is rain Water harvesting? What is then need for RHW? (10)

(d) How dose acid rain form and what is the effect of acid rain? (10)

5. (a) Define pollution. State different types of pollution. (10)

(b) Solid waste management from industrial and household sources is a big challenge – discuss. (10)

Or

(c) Mention any five social issues relating to environment and briefly. Mention the effect. (10)

(d) Name the frequently accruing natural disaster in India and dwell on the impact of one of them. (10)

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CP–6296

Sub. Code
51

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

COMPUTER PROGRAMMING AND UTILITIES

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

1. (a) Describe the generations of computer. (10)

(b) Explain any one of the output devices. (5)

Or

(c) Describe the classifications of computer. (10)

(d) Explain the role of control unit in a computer. (5)

2. (a) List out the components of database and explain. (5)

(b) What is data dictionary? Explain. (5)

(c) Briefly explain the concepts of RDBMS. (5)

Or

(d) Discuss the need for database. (5)

(e) List out the bata types and give example for each one of them. (5)

(f) Explain the elements of Relational model. (5)

3. (a) Define the following ad give example for each one of them.

   (i) Identifier

   (ii) Keyword

   (iii) Constant

   (iv) Variable

   (v) Program. (10)

   (b) What is a function? Give an example. (5)

Or

   (c) Give the general syntax of for statement and explain its usage with suitable example. (8)

   (d) What is an array? Write a 'C' program with the usage of an array. (7)

4. (a) Explain the different network topologies. (10)

   (b) Write a note on internet security. (5)

Or

   (c) Explain OSI Reference model. (10)

   (d) Write a note on extranet. (5)

5. (a) List any five mathematical functions and any five financial functions. Explain each one with example. (10)

   (b) How do you create a bar chart? Explain. (5)

Or

   (c) Explain any five spread sheet commands. (5)

   (d) How do you create a pie chart? Explain. (5)

   (e) Write any five statistical functions with suitable example. (5)

CP–6297

Sub. Code
52

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

NAVIGATION – V

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Explain the properties of a free gyroscope. (8)

   (b) List the operational errors of a gyro compass. (7)

   Or

   (c) With suitable sketch explain the construction of a gyro compass. (15)

2. (a) Draw block diagram of radar and explain its part, in detail. (15)

   Or

   (b) Explain :
   (i) Pulse length
   (ii) HBW
   (iii) VBW
   (iv) Gain
   (v) Anti – Rain clutter in detail. (15)

3. (a) What are the inputs of a course Recorder? (5)

   (b) How does the course recorder work? (5)

   (c) Explain the checks carried out and care taken on the course recorder. (5)

   Or

(d) What is the principle of an Auto – Pilot? (5)

(e) Explain the controls of an Auto – Pilot. (5)

(f) Explain OFF course alarm and its limitations of an Auto – Pilot. (5)

4. (a) Carry out a neat radar plot using large scale and compile a full report of (15)

(i) CPA

(ii) TCPA course and speed of target

(iii) Aspect at 0518 hr.

Own course –340° (T) speed 12 kts.

Ships time	Bearing (T)	Range of Target (miles)
0509	285.5	4.4
0512	280.0	3.5
0518	253.0	1.6

Or

(b) Own ship's course 180° (T) speed 6 kts. Following observations were made of a target. (15)

Ships Time	Bearing (T)	Range (M)
2104	030	6.5
2116	036	4.6

Find the following :

(i) CPA and TCPA

(ii) Course and speed of target

(iii) Aspect at 2116.

5. (a) Explain all the various Alarms on the bridge. (15)

Or

(b) State, in detail, the action to be taken on hearing the fire detector alarm. (15)

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CP–6298

Sub. Code
53

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

VOYAGE PLANNING COLLISION PREVENTION AND MARINE COMMUNICATION V

Time : 2 Hours Maximum : 25 Marks

Answer **all** questions.

(5 × 5 = 25)

1. (a) What are the different factors that forced the invention /introduction of GMDSS. (5)

Or

(b) (i) Write short notes on radio wave propagation. (2)

(ii) How do you restrict the range of radio wave travel? (3)

2. (a) (i) What is the basic 'system concept' of IAMSAR communication? (3)

(ii) Explain safety net. (2)

Or

(b) Explain the following : (5)

(i) MMSI (ii) MRCC (iii) MSI (iv) NBDP (v) PIW (vi) INMARSAT (vii) interco (viii) GMDSS (ix) EPIPB (x) SART.

3. (a) What is the arrangement to achieve WWNS? Which area is assigned to India and how it is operated? (5)

Or

(b) What is bridge resource management? What are the benefits of BRM? (5)

4. (a) Find depth of water, when a vessel drawing 6.4 mtr will have like when crossing 5.4 mtr patch at vancouer harbour on 12 may at 1400 hrs standard time. (local time)

Time zone + 0800

Time	–	Ht of tide
0445	–	4.4 mtr
1120	–	0.3 mtr
1810	–	4.3 mtr
2315	–	3.3 mtr.

Or

(b) Write short notes on : (5)

(i) Tide (ii) range of tide (iii) chart datum (iv) bore tide (v) neap tide.

5. (a) Find the earliest time on 17th June, A vessel off Boston, drawing 3.4 mtr will be able to cross the bar marked 2 mtr with a clearance of 1 mtr. under the keel. att extracts : (5)

	Time	Height
16th	0229	= 3.1 mtr
	0848	(–) 0.1 mtr (minus)
	1508	= 2.9 mtr
	2109	= 0.2 mtr
17th	0325	= 0.3 mtr
	0939	= 0.00 mtr
	1601	= 2.8 mtr
	2205	= 0.3 mtr

Or

(b) Draw lights and shapes displayed Bg :

(i) Pilot V/L

(ii) Fishing V/L

(iii) V/L engaged in fishing

(iv) Dredger

(v) Towing V/L

(vi) Ram

(vii) Sailing V/L

(viii) Trawlers (all cases).

CP–6299

Sub. Code
54

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

SHIPMASTERS BUSINESS

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions.

(5 × 20 = 100)

1. (a) Mention various documents to be carried on board. (10)

   (b) Explain the purpose and benefits of ship registration. (10)

   Or

   (c) Master has a role to play in every aspect of shipping. Discuss. (20)

2. (a) How does a ship get health clearance for entering harbour? (10)

   (b) What type of behaviour can be constructed as endangering a ship? (10)

   Or

   (c) Explain the Custom House procedure for entering and leaving harbour. (20)

3. (a) Define : Distress and salvage. (10)

   (b) How should a peril be handled by a ship? (10)

   Or

   (c) How does "SAR" organisation work? (20)

4. (a) Explain the following : (20)

   (i) Notice of readiness

   (ii) Quarantine anchorage

   (iii) Article of agreement

   (iv) Note of protests.

Or

   (b) What are duties of shipping master? (10)

   (c) How does multimodel transportation help improve cargo movement? (10)

5. (a) How does a vessel gets fixed to carry particular cargo? (10)

   (b) Explain following terms :

   Weather working days and Sundays and holidays excepted. (10)

Or

   (c) Explain the importance of "Mates receipt" and Bills of Lading. (20)

CP–6300

Sub. Code
55

B.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Nautical Science

MARINE MANAGEMENT AND COMMERCE

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions.

(5 × 20 = 100)

1. (a) Give an account on the formal safety management.

   (b) Explain in detail about the human resource planning.

   Or

   (c) Give a note on the importance of statistics on marine management.

   (d) Explain in detail on the Indian fleet.

2. (a) Discuss in details on the problems of ballast water and its preventive measures.

   (b) Write a note on the cargo management.

   Or

   (c) Give an account on port's function and its range of service.

   (d) Write a short note on maritime fraud.

3. (a) What are the roles of intermediaries in shipping business?

(b) Give an account on the different agents involved in shipping.

Or

(c) Explain in detail about the liner traders.

(d) What are the roles of freight broker in shipping industry?

4. (a) How the mates receipt is used in the bill lading?

(b) Give an account on the multi modal transport.

Or

(c) Write short notes on the uses of chartering.

(d) Give an account on the zone problems in load line calculations.

5. (a) How to manage (conserve) the marine environment for the sustainable development?

(b) What are the steps involved in the operation management?

Or

(c) Explain in detail about the evolution of world merchant fleet.

(d) Give an account on the shipping services.